

Erosion of Safety Culture at Boeing

Case Study Analysis

BUS-7200 Value Systems and Professional Ethics

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Guided by: Prof. Michael Deardorff

Group Members:

Amruthavalli Palluri

Asa Mpuan

Patrick Brown

FNU Rahul

Sagar Lamsal

Introduction

The Boeing 737 MAX crisis represents one of the most significant ethical failures in modern corporate and engineering history. Two fatal crashes in 2018 and 2019 resulted in the deaths of 346 individuals and exposed systemic weaknesses in corporate governance, regulatory oversight, and organizational culture. Subsequent reviews by the Federal Aviation Administration (FAA) and the U.S. House Committee on Transportation and Infrastructure identified deficiencies in certification oversight, risk communication, and organizational decision-making processes (FAA, 2022; U.S. House Committee on Transportation and Infrastructure, 2020). Separately, the U.S. Department of Justice (DOJ) examined Boeing's representations to regulators in connection with the aircraft's certification.

In July 2024, the DOJ announced that Boeing had reached an agreement in principle to resolve a criminal charge related to certification representations; however, the proposed plea agreement was later rejected by a federal judge in December 2024, reflecting continued judicial scrutiny (U.S. Department of Justice [DOJ], 2024). In November 2025, the court approved the DOJ's motion to dismiss the criminal case under a revised agreement, a development that generated controversy among victims' families and observers (Reuters, 2025). These developments underscore the gravity of ethical, legal, and governance breakdowns within the organization. The Boeing case provides a powerful framework for examining the interaction of professional ethics, corporate governance, technological complexity, and stakeholder responsibility in safety-critical industries.

Case Identification and Organizational Context

Historically, Boeing was widely recognized as an engineering-driven organization that prioritized safety, technical excellence, and conservative design philosophy. However, over the past several decades, analysts and scholars have noted a gradual cultural shift following corporate restructuring and leadership transitions that increasingly emphasized cost control, shareholder value, and competitive positioning (Mintz & Miller, 2025). This shift did not occur abruptly. Rather, it emerged incrementally through incentive structures that rewarded production speed, financial performance, and market competitiveness more strongly than technical prudence and system redundancy.

The redesign of the 737 platform into the 737 MAX illustrates how competitive pressure shaped decision-making. In response to Airbus's A320neo, Boeing opted to update an existing airframe rather than develop an entirely new aircraft. Larger engines were repositioned further forward on the wings, altering aerodynamic characteristics and creating a tendency for the aircraft's nose to pitch upward under certain conditions (U.S. House Committee on Transportation and Infrastructure, 2020). To address this issue, Boeing implemented the Maneuvering Characteristics Augmentation System (MCAS), an automated flight control software intended to adjust the aircraft's nose position. However, MCAS relied on input from a single angle-of-attack sensor, creating a critical single point of failure (Federal Aviation Administration [FAA], 2022). When erroneous sensor data triggered MCAS repeatedly, pilots encountered automated system behavior they had not been fully informed about or trained to manage.

The ethical dimension deepened because pilots were not fully informed of MCAS functionality, nor was simulator training universally mandated. Investigations later revealed that certain system characteristics were not sufficiently disclosed during certification processes (U.S. Department of Justice [DOJ], 2024). The result was not simply a software malfunction but a breakdown in transparency, risk management, and professional responsibility.

Stakeholder Implications

The Boeing case implicates multiple stakeholder groups. Passengers possessed a fundamental right to safety and informed consent regarding aircraft systems. Pilots and flight crews had the right to accurate operational manuals, comprehensive disclosure of aircraft systems, and appropriate training. Engineers within Boeing held professional obligations to prioritize public safety in accordance with established engineering ethics principles (Herkert et al., 2020). Senior executives and board members bore fiduciary duties not only to shareholders but to all stakeholders whose lives and livelihoods depended upon safe aircraft manufacturing. The FAA, as the primary regulator, carries the responsibility of independent oversight to ensure compliance with safety standards. These intersecting responsibilities illustrate how failures in governance affect multiple constituencies simultaneously (Mintz & Miller, 2025).

The Organizational Designation Authorization (ODA) structure permitted Boeing employees to act on behalf of the FAA in certain certification processes. While this delegation was designed to improve administrative efficiency, congressional investigations raised concerns about weakened oversight and the concentration of certification authority within the manufacturer (U.S. House Committee on Transportation and Infrastructure, 2020). The

intersection of delegated authority, production pressure, and financial incentives created a structural environment vulnerable to ethical erosion.

Ethical Framework Analysis

The central ethical dilemma confronting Boeing leadership can be framed as a conflict between competitive survival and uncompromising safety assurance. Under intense market pressure from Airbus, leadership faced incentives to accelerate certification and avoid costly training requirements that might disadvantage the aircraft commercially. The ethical question becomes whether short-term financial and competitive benefits can ever justify incremental safety risk in an industry where failures result in catastrophic loss of life (Federal Aviation Administration [FAA], 2022; U.S. House Committee on Transportation and Infrastructure, 2020).

From a deontological perspective, the answer is unequivocal. Duty-based ethics asserts that human life must never be treated as a variable within a cost-benefit calculation. Boeing had an absolute moral obligation to ensure that identifiable safety risks were thoroughly mitigated prior to aircraft deployment (Herkert et al., 2020). This duty extends beyond regulatory minimum compliance to proactive risk mitigation.

A narrow utilitarian analysis might initially suggest that preserving jobs, market share, and economic stability could justify expedited production. However, a comprehensive utilitarian framework requires the inclusion of all stakeholders and all consequences. Once catastrophic loss of life, global reputational damage, criminal penalties, and long-term

economic loss are incorporated into the calculation, the utilitarian case collapses (FAA, 2022). The aggregate harm vastly exceeded any short-term economic benefit.

Rights theory further clarifies the ethical violation. Passengers and pilots had the right to full disclosure of system behavior affecting flight control (U.S. Department of Justice [DOJ], 2024). Withholding material information about MCAS functionality undermined informed consent and violated these rights. From a virtue ethics standpoint, the case reflects deficiencies in organizational courage, prudence, and transparency. Ethical organizations cultivate cultures where dissent is encouraged, and safety concerns are elevated rather than suppressed.

Technology, Automation Bias, and Information Asymmetry

The role of technology in the crisis highlights the interaction between software systems and organizational culture. Automation bias refers to the tendency to over-rely on automated systems under the assumption that algorithmic intervention enhances safety (Herkert et al., 2020). In the 737 MAX case, excessive confidence in MCAS performance contributed to the underestimation of failure risk (Federal Aviation Administration [FAA], 2022; U.S. House Committee on Transportation and Infrastructure, 2020). Moreover, the reliance on a single angle-of-attack sensor contradicted long-standing aviation redundancy principles (FAA, 2022).

Information asymmetry compounded the problem. Pilots were placed in situations where automated override behavior occurred without full knowledge of system design logic (U.S. Department of Justice [DOJ], 2024). Ethical system implementation requires that end users possess adequate information to respond to risk scenarios. Technology does not operate

independently of ethical judgment; rather, it reflects managerial choices regarding disclosure, redundancy, and risk tolerance.

Evaluation of Alternative Courses of Action

One potential course of action available to Boeing leadership was to maintain production schedules while incrementally addressing safety concerns. This profit-centric approach offered short-term advantages, including preservation of cash flow, avoidance of contractual penalties, and maintenance of shareholder confidence. However, it sustained systemic risk exposure and increased the likelihood of further catastrophic failure. Subsequent regulatory reviews and investigations highlighted the extensive financial, operational, and reputational consequences that followed the crashes (Federal Aviation Administration [FAA], 2022).

An alternative course of action involved halting production to conduct a comprehensive engineering-driven safety reset. This approach would include redesigning MCAS with sensor redundancy, mandating simulator training, separating safety reporting lines from production management, and fully cooperating with regulators. Although this strategy entails short-term financial losses and market vulnerability, it addresses root causes rather than surface symptoms (FAA, 2022; U.S. Department of Justice [DOJ], 2024).

A comparative analysis suggests that the engineering-driven safety reset is ethically superior. It satisfies deontological duty by prioritizing human life, fulfills stakeholder

obligations by restoring transparency, and produces stronger long-term utilitarian outcomes by reducing systemic risk and preserving institutional trust (Mintz & Miller, 2025).

Selected Course of Action and Justification

The most ethically defensible and strategically sustainable course of action is a comprehensive engineering-driven safety reset supported by enhanced transparency and strengthened regulatory collaboration (Federal Aviation Administration [FAA], 2022; U.S. Department of Justice [DOJ], 2024). This approach requires not merely technical correction but cultural transformation. Safety authority should be structurally independent of production targets, and engineers must be empowered to raise concerns without retaliation. Executive compensation structures should incorporate safety performance metrics alongside financial indicators to realign organizational incentives (Mintz & Miller, 2025).

The short-term financial drawbacks of this strategy are significant. Production delays, contractual penalties, and temporary reputational exposure are likely consequences of comprehensive reform (FAA, 2022). However, financial losses are recoverable through operational improvement and restored market confidence. The loss of human life and erosion of public trust are not reversible. Accordingly, ethical analysis supports the conclusion that the benefits of comprehensive reform outweigh the costs.

Governance and Cultural Reform

The deeper failure in the Boeing case reflects weaknesses in governance architecture. Effective oversight in safety-critical industries requires board-level expertise capable of

challenging technical assumptions and scrutinizing safety-related decisions (Mintz & Miller, 2025). Incentive structures that disproportionately reward stock performance and delivery volume can create moral hazard by distorting organizational priorities away from risk mitigation (Mintz & Miller, 2025). Reforms should therefore include stronger board engagement in safety oversight, independent safety reporting channels, and reevaluation of delegated regulatory authority structures that concentrate certification authority within the manufacturer (U.S. House Committee on Transportation and Infrastructure, 2020).

Cultural reconstruction is equally critical. Ethical erosion occurs gradually when organizational silence replaces professional dissent and when performance pressures discourage the escalation of safety concerns (Mintz & Miller, 2025). Establishing a formal “Safety Stop” authority, whereby any engineer can halt production without fear of retaliation, reinforces psychological safety and ethical courage. Transparency with regulators and the public should be institutionalized as a permanent operating principle rather than deployed only in response to crisis (Federal Aviation Administration [FAA], 2022).

Long-Term Ethical Implications

Organizational trust functions as a critical intangible asset in safety-critical industries. Airlines and passengers depend on sustained confidence in engineering integrity and regulatory compliance. Research in corporate governance suggests that organizations with strong ethical cultures experience greater long-term resilience, reduced litigation exposure, and stronger stakeholder loyalty (Mintz & Miller, 2025). The 737 MAX crisis illustrates that ethical compromise is not only morally indefensible but also strategically unsustainable.

In safety-critical industries, legitimacy derives from demonstrated commitment to protecting human life. When leadership allows financial metrics to eclipse moral responsibility, institutional fragility emerges and long-term viability is threatened (Mintz & Miller, 2025). The Boeing case therefore serves as a cautionary example for technology-driven organizations operating under competitive pressure.

Conclusion

The erosion of safety culture at Boeing reflects the convergence of competitive pressure, automation overconfidence, regulatory weakness, and misaligned incentives. The profit-centric strategy was ethically flawed and strategically shortsighted. A comprehensive engineering-driven safety reset grounded in transparency, stakeholder responsibility, and governance reform represents the only defensible path forward (Mintz & Miller, 2025).

In aviation, safety is not a negotiable variable within financial calculus. It is the foundational principle upon which corporate legitimacy rests. Sustainable success requires deliberate reinforcement of engineering integrity, transparent leadership, independent oversight, and unwavering commitment to human life. The Boeing case ultimately demonstrates that ethical governance is not merely a compliance requirement but a strategic necessity in safety-critical industries.

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